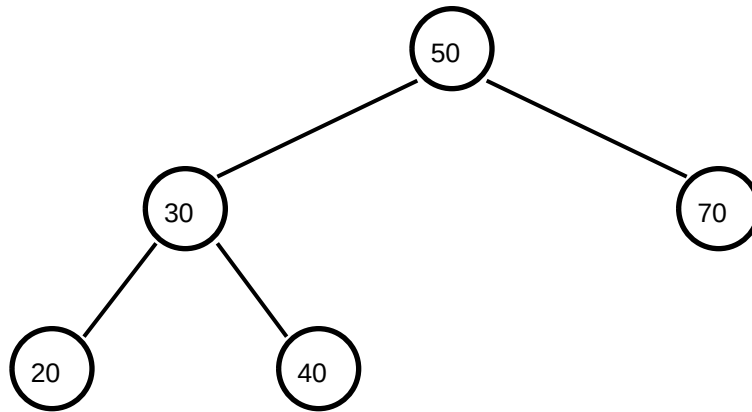


# Computer Science Exam

## Q1. Describe the structure of a binary search tree.

Consider the following tree and answer parts (a) and (b).

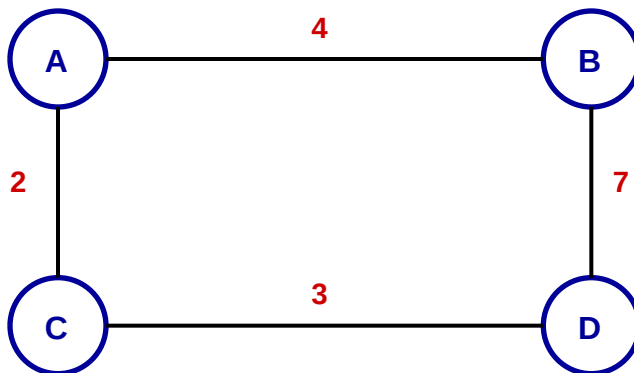


(a) What is the in-order traversal of this tree?

(b) Insert the value 35 into this BST.

## Q2. Explain Dijkstra's algorithm with an example.

Use the following weighted graph to find the shortest path from A to D.



**Q3. Define the following terms:**

- (a) Time complexity
- (b) Space complexity
- (c) Big-O notation